Chapter 1 Notes

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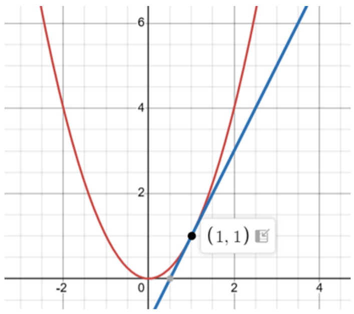
1 Functional Derivatives

Recall that the derivative from first principles is defined as

$$\frac{df(x)}{dx} = \lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}.$$

The interpretation of this definition is a graphical one. It is the instantaneous slope at a point on a graph.



Analogously, there exists the functional derivative. By definition, it is

$$\frac{\delta F[f(x')]}{\delta f(x)} = \lim_{\epsilon \to 0} \frac{F[f(x') + \epsilon\delta(x - x')] - F[f(x')]}{\epsilon}.$$

x' is a dummy variable, typically integrated out. $\epsilon\delta(x - x')$ can be considered a test "function" that only really makes sense in an integral. The Dirac delta slightly perturbs the original input $f(x')$ at a specific location $x = x'$, resulting in an output dependent on x. Functional derivatives are a bit trickier than ordinary derivatives so it's best to put them into practice.

Given $H[f] = \int_a^b g[f(x)]\, dx$,

$$\frac{\delta H[f(x)]}{\delta f(x_0)} = \lim_{\epsilon \to 0} \frac{1}{\epsilon}\Big[\int_a^b g[f(x) + \epsilon\delta(x_0 - x)]\, dx - \int_a^b g[f(x)]\, dx\Big]$$

$$\lim_{\epsilon \to 0} \frac{1}{\epsilon} \int_a^b \epsilon \frac{dg[f(x)]}{df(x)} \delta(x_0 - x)\, dx$$

$$= \frac{dg[f(x_0]}{df(x_0)}$$

where we expanded the Taylor series in ϵ for $g[f(x) + \epsilon\delta(x_0 - x)]$.

Likewise, if $J[f] = \int g(f')\, dy$ with $f' = \frac{df}{dy}$, $\frac{\delta J[f(y)]}{\delta f(x)} = -\frac{d}{dx}(\frac{dg[f'(x)]}{df'(x)})$.

2 Inspiring the Lagrangian

Average kinetic energy and average potential energy can be expressed as $\bar{T} = \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} \frac{1}{2} m[\dot{x}(t)]^2\, dt$ and $\bar{V} = \frac{1}{\mathcal{T}} \int_o^{\mathcal{T}} V[x(t)]\, dt$, respectively, where $\mathcal{T}$ is the time interval over some trajectory. From the previous section, we may apply the functional derivatives to these two quantities to discover $\frac{\delta \bar{V}[x(t)]}{\delta x(t)} = \frac{1}{\mathcal{T}} \frac{dV}{dx}$ and $\frac{\delta \bar{T}[x(t)]}{\delta x(t)} = \frac{m\ddot{x}}{\mathcal{T}}$. These are the rate of change of $\bar{V}$ and $\bar{T}$ over a variation in the path $x(t)$. Ignoring the prefactor $\frac{1}{\mathcal{T}}$ and equating the two quantities, behold, Newton's second law:

$$m\ddot{x} = -\frac{dV}{dx}.$$

Rearranging, it is equivalent to say

$$\frac{\delta}{\delta x(t)}(\bar{T}[x] - \bar{V}[x]) = 0.$$

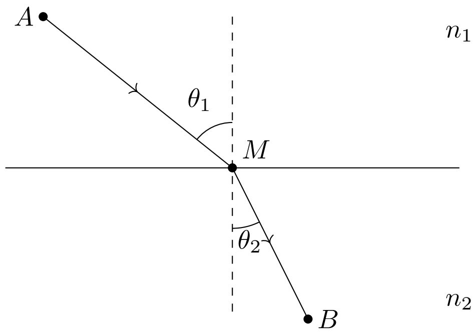
The Lagrangian $L = T - V$ is motivated. The action is defined as $S = \int_0^{\mathcal{T}} L\, dt$. Therefore,

$$\frac{\delta S}{\delta x(t)} = 0,$$

which is known as Hamilton's principle of least action. Lot to unpack here.

The action is a quantity that can be minimized through a functional derivative which involves a variation in paths. When you minimize it, it is equivalent to rederiving Newton's second law from introductory physics which we know is true. This can be further illustrated by light travelling through two different media.

Consider a light source that starts in the first medium at point A and has to reach point B in the second medium. For the most part, the light will travel through a straight line except when it bends at the interface due to Snell's law. Curiously, from point A to point B, the light will always pick the path of least time, which can rederive Snell's law. This is known as Fermat's principle.



The connection is that Snell's law is like Newton's second law. This is a classical picture where cause and effect are responsible for the underlying physics. However, we can equivalently see that Fermat's principle is like Hamilton's principle of least action. This is a global picture where there appears to be some quantity defined throughout the entire path that must be minimized beforehand. Both of the physics work, but the Lagrangian, inspired by the global picture, will prove to be a much stronger tool in tackling more complicated problems like QFT.

3 Phase Factor

There is one more connection to be drawn between light rays and the principle of least action. Light rays have phasors e^{ikx} where the phase depends on the path length. If light rays from multiple paths were to converge on a single point, the brightness at that point is determined by how in phase all the sources are impinging on the point. If there is constructive interference, the point would be bright.

That same constructive interference can be extended to the principle of least action with each possible path taken (variations in path) acquiring a phase $e^{iS/\hbar}$. If the probability amplitude sums the phases of all possible paths, only paths close to the true trajectory– obeying the equations of motion– will constructively interfere and produce the observed, physical path.

Practice Questions

1. Euler-Lagrange Equation

The Lagrangian density $\mathcal{L}$ can be defined as $L = \int d^3x\ \mathcal{L}$. Minimizing the action S from first principles of the functional derivative with $\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi)$, show that

$$\frac{\partial \mathcal{L}}{\partial \phi} = \partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \Big).$$

2. Circle of Light

Imagine a detector located in the exact center of a circle. There are N points equally spaced around the circle that emit light in a peculiar way. The first point emits light with an amplitude $\cos(\omega t)$ with each subsequent adjacent point counterclockwise around the circle having a phase shift of $2\pi/N$ (so the next point would have amplitude $\cos(\omega t + 2\pi/N)$).

If we count all N point sources contributing their amplitude towards the center, show that the intensity the detector reads is null.